

5 Some fundamental distributions

Distributions such as the Bernoulli and binomial distributions occur naturally in probability theory and serve as building blocks in probabilistic modelling. In this section we introduce several discrete and continuous univariate distributions which should be in the toolbox of every budding probabilist. For each distribution we give both the mathematical definition and its motivation, thereby introducing some fundamental probabilistic ideas and techniques. In each case we assume that an appropriate probability space $(\Omega, \mathbb{P})$ has been given (see Remark 4.37). Several other important distributions will be introduced throughout the course.

5.1 Multinomial distribution

The *multinomial distribution* generalizes the binomial distribution to trials with multiple outcomes. To define it we need to generalize the binomial coefficient. For $n \geq 1, K \geq 1$ and non-negative integers $n_1, \dots, n_K$ satisfying $n_1 + \dots + n_K = n$, we define the *multinomial coefficient* $\binom{n}{n_1, n_2, \dots, n_K}$ by

$$\binom{n}{n_1, n_2, \dots, n_K} = \frac{n!}{n_1! n_2! \dots n_K!}. \quad (5.1)$$

For example, we have $\binom{6}{1, 2, 4} = \frac{6!}{1! 2! 4!} = 15$.

Definition 5.1 (Multinomial distribution). *Let $n \geq 1, K \geq 2$ be integers and $\mathbf{p} = (p_1, \dots, p_K)$ be a vector of non-negative real numbers satisfying $p_1 + \dots + p_K = 1$.¹ A discrete K -dimensional random vector $\mathbf{X} = (X_1, \dots, X_K)$ is said to follow the Multinomial($n, \mathbf{p}$) distribution, written $\mathbf{X} \sim \text{Multinomial}(n, \mathbf{p})$, if its probability mass function is given by*

$$f_{\mathbf{X}}(\mathbf{x}) = \binom{n}{x_1, \dots, x_K} p_1^{x_1} \dots p_K^{x_K}, \quad (5.2)$$

for $\mathbf{x} = (x_1, \dots, x_K) \in \mathbb{Z}_+^K$ satisfying $\sum_{i=1}^K x_i = n$, and $f_{\mathbf{X}}(\mathbf{x}) = 0$ otherwise.

Intuitively, the components $X_1, \dots, X_K$ of $\mathbf{X}$ count the frequencies in n independent trials with K possible (distinct) outcomes, say $c_1, \dots, c_K$. Let's express this idea precisely. Given $n \geq 1$ and $\mathbf{p} = (p_1, \dots, p_K)$, let $Y_1, \dots, Y_n$ be i.i.d. random variables such that

$$\mathbb{P}(Y_i = c_k) = p_k, \quad k = 1, \dots, K.$$

Thus Y_i is the outcome of the i -th trial.

Remark 5.2 (Categorical distribution). We may think of $c_1, \dots, c_K$ as K *categories* or *classes* which classify the outcome of the trial. The distribution of Y_i is said to be *categorical*. It may be regarded as a generalization of the Bernoulli distribution (where $K = 2, c_1 = 0$ and $c_2 = 1$).

¹Thus $\mathbf{p}$ is a *probability vector* with K components.

Let $Z_{ik} = I_{\{Y_i = c_k\}}$ be the indicator that outcome of the i -th trial is c_k , and define $\mathbf{Z}_i = (Z_{i1}, \dots, Z_{iK})$ which is a random vector. For example, suppose $K = 3$. Then $\mathbf{Z}_i = (0, 1, 0)$ means that the i -th trial results in outcome y_2 . Let

$$\mathbf{X} = (X_1, \dots, X_K) = \mathbf{Z}_1 + \dots + \mathbf{Z}_n, \quad (5.3)$$

where $X_k = \sum_{i=1}^n Z_{ik}$, be the vector of counts.

Theorem 5.3. *Under the setting above we have $\mathbf{X} \sim \text{Mutlinomial}(n, \mathbf{p})$.*

The proof is very similar to that of Theorem 4.38 (where $K = 2$). We first give a combinatorial interpretation of the multinomial coefficient. Recall the notation $[n] = \{1, 2, \dots, n\}$.

Lemma 5.4. *Let $n \geq 1$, $K \geq 1$, and let $n_1, \dots, n_K$ be non-negative integers such that $n_1 + \dots + n_K = n$. Consider the set*

$$\mathcal{P}_{n_1, \dots, n_K} = \{(A_1, \dots, A_K) : A_k \subset [n] \text{ pairwise disjoint, } |A_k| = n_k\}.$$

Then $|\mathcal{P}_{n_1, \dots, n_K}| = \binom{n}{n_1, \dots, n_K}$.

Proof. We think of the sets $A_1, \dots, A_K$ as a partition of $[n]$ with given sizes $n_1, \dots, n_K$. (Note that when $K = 2$ we have $A_2 = A_1^c$. So this reduces to the situation in 3.11(ii).) To pick such a partition, we first pick n_1 elements in $[n]$ to form A_1 , and there are $\binom{n}{n_1}$ ways. Next we pick n_2 elements from the remaining $n - n_1$ elements to form A_2 , and there are $\binom{n - n_1}{n_2}$ ways. In the last step, there are $n - n_1 - \dots - n_{K-1} = n_K$ elements which will form A_K . So we have, by the product rule and thanks to the cancellations,

$$\begin{aligned} |\mathcal{P}_{n_1, \dots, n_K}| &= \binom{n}{n_1} \binom{n - n_1}{n_2} \dots \binom{n - n_1 - \dots - n_{K-1}}{n_K} \\ &= \frac{n!}{n_1!(n - n_1)!} \frac{(n - n_1)!}{n_2!(n - n_1 - n_2)!} \dots \frac{(n - n_1 - \dots - n_{K-1})!}{n_K!0!} \\ &= \frac{n!}{n_1! \dots n_K!} = \binom{n}{n_1, \dots, n_K}. \end{aligned}$$

□

Proof of Theorem 5.3. The joint pmf of $Y_1, \dots, Y_n$ is given by

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = \prod_{i=1}^n \left(\prod_{k=1}^K p_k^{I_{\{y_i = c_k\}}} \right) = \prod_{k=1}^K p_k^{\sum_{i=1}^n I_{\{y_i = c_k\}}},$$

where $\sum_{i=1}^n I_{\{y_i = c_k\}}$ is the number of times for which $y_i = c_k$. Let $\mathbf{x} = (x_1, \dots, x_K) \in \mathbb{Z}_+^K$ where $\sum_{k=1}^K x_k = n$. (It is clear that otherwise we must have $\mathbb{P}(\mathbf{X} = \mathbf{x}) = 0$). We have

$$\mathbb{P}(\mathbf{X} = \mathbf{x}) = \sum_{(y_1, \dots, y_n) : \sum_{i=1}^n I_{\{y_i = c_k\}} = x_k} \prod_{k=1}^K p_k^{x_k} = \binom{n}{x_1, \dots, x_K} p_1^{x_1} \dots p_K^{x_K},$$

where the first equality follows from the sum rule and the second equality follows from the previous lemma. Thus $\mathbf{X} \sim \text{Mutlinomial}(n, \mathbf{p})$. □

5.2 Geometric distribution

Definition 5.5 (Geometric distribution). *Let $0 < p < 1$. A discrete real-valued random variable X is said to have the Geometric(p) distribution, written $X \sim \text{Geometric}(p)$, if its probability mass function is given by*

$$f_X(x) = \begin{cases} (1-p)^{x-1}p, & \text{for } x \in \{1, 2, \dots\}; \\ 0, & \text{otherwise.} \end{cases} \quad (5.4)$$

The name *geometric distribution* can be motivated by its relation with the geometric series. By the *geometric sum formula* (see (2.3)), we have

$$\sum_{x=1}^{\infty} f_X(x) = \sum_{x=1}^{\infty} (1-p)^{x-1} p \stackrel{i=x-1}{=} p \sum_{i=0}^{\infty} (1-p)^i = \frac{p}{1-(1-p)} = 1. \quad (5.5)$$

So the normalization condition (2.17) is satisfied.

To motivate probabilistically the geometric distribution, let $X_1, X_2, \dots$ be an infinite sequence of i.i.d. Bernoulli(p) random variables.² Introduce a random variable T , with values in $\{1, 2, \dots\} \cup \{\infty\}$, defined by

$$T(\omega) = \begin{cases} \min\{n : X_n(\omega) = 1\}, & \text{if } X_n(\omega) = 1 \text{ for some } n; \\ \infty, & \text{otherwise.} \end{cases} \quad (5.6)$$

Intuitively, T represents the *first time* that we observe a success (represented by 1) in the sequence of Bernoulli trials. If all $X_n(\omega) = 0$, then $T(\omega) = \infty$. The concept of *time*, whose *directional* nature is of profound practical relevance, plays a key role in further studies in probability theory such as *stochastic processes* and their applications.

Before stating the result we introduce a useful definition.

Definition 5.6 (Almost sure). *Given a probability space $(\Omega, \mathbb{P})$, an event A is said to occur almost surely (a.s.) if $\mathbb{P}(A) = 1$.*

Theorem 5.7 (Geometric distribution from an i.i.d. sequence of Bernoulli random variable). *Consider the random variable T defined by (5.6). Then $T \sim \text{Geometric}(p)$. In particular, we have $T < \infty$ almost surely.*

Proof. We compute the pmf of T from the definition (5.6). We start with $\mathbb{P}(T = 1)$. Since $\{T = 1\} = \{X_1 = 1\}$, we have

$$\mathbb{P}(T = 1) = \mathbb{P}(X_1 = 1) = p.$$

Next let t be a positive integer with $n \geq 2$. Then

$$\{T = t\} = \{X_1 = 0, \dots, X_{t-1} = 0, X_t = 1\}.$$
³

²Recall Definition 4.40.

³Here we use t instead of x since the symbol X_x , while mathematically correct, may look confusing to some students. It also looks more intuitive since t is interpreted here as time.

By the joint independence of $X_1, \dots, X_t$, we have

$$\begin{aligned}\mathbb{P}(T = t) &= \mathbb{P}(X_1 = 0, \dots, X_{t-1} = 0, X_t = 1) \\ &= \mathbb{P}(X_1 = 0) \cdots \mathbb{P}(X_{t-1} = 0) \mathbb{P}(X_t = 1) \\ &= (1 - p)^{t-1} p.\end{aligned}$$

From (5.5), we have

$$\mathbb{P}(T = \infty) = 1 - \sum_{t=1}^{\infty} \mathbb{P}(T = t) = 1 - \sum_{t=1}^{\infty} (1 - p)^{t-1} p = 0.$$

Hence T is almost surely finite, i.e., $\mathbb{P}(T < \infty) = 1$, and its pmf is given by (5.6). $\square$

Remark 5.8 (Another representation of the geometric distribution). Sometimes it is convenient to motivate the geometric distribution in terms of the number of *failures* before the first success. As an exercise, you should verify that in this case the probability mass function is given by

$$f_X(x) = \begin{cases} (1 - p)^x p, & \text{for } x \in \{0, 1, \dots\}; \\ 0, & \text{otherwise.} \end{cases}$$

Unless otherwise specified we will use the convention in Definition 5.5.

We note a special property satisfied by the geometric distribution.

Proposition 5.9 (Memoryless property). *Let $0 < p < 1$ and suppose $T \sim \text{Geometric}(p)$. Then, for any positive integers s and t we have*

$$\mathbb{P}(T = s + t | T > s) = \mathbb{P}(T = t). \quad (5.7)$$

Proof. By the definition of conditional probability, we have

$$\begin{aligned}\mathbb{P}(T = s + t | T > s) &= \frac{\mathbb{P}(T = s + t \text{ and } T > s)}{\mathbb{P}(T > s)} \\ &= \frac{\mathbb{P}(T = s + t)}{\mathbb{P}(T > s)} \quad (\text{since } T = s + t \text{ implies } T > s) \\ &= \frac{(1 - p)^{s+t-1} p}{(1 - p)^s} \\ &= (1 - p)^{t-1} p = \mathbb{P}(T = t).\end{aligned}$$

The third equality uses the fact that the tail probability of the geometric distribution is given by

$$\mathbb{P}(T > s) = (1 - p)^s. \quad (5.8)$$

(See Exercise 1 in Section A.5.1.) $\square$

Why is this called the memoryless property? Suppose T represents the waiting time of a bus measured in minutes (say). The event $T > s$ means that you have waited for s minutes and the bus has not arrived yet. Now (5.7) means the following. Given the event $T > s$, the conditional distribution of the *additional* waiting time $T - s$ does not depend on the time you have spent and is equal to $\text{Geometric}(p)$, which is the unconditional distribution of T . In this sense, the waiting time does not remember how long you have waited.

5.3 Poisson distribution

Recall the *exponential function* which can be defined in terms of a *power series* by

$$\exp(x) = e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \quad x \in \mathbb{R}. \quad (5.9)$$

Let $x = \lambda > 0$. Dividing both sides of (5.9) by e^λ , we have

$$\sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = 1. \quad (5.10)$$

Equivalently, e^x can be defined by the limit

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n. \quad (5.11)$$

More generally, one can show that whenever $\lim_{n \rightarrow \infty} x_n = x$, we have

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x_n}{n}\right)^n. \quad (5.12)$$

Mathematically, the (pmf of the) *Poisson distribution* corresponds to the terms of the left hand side of (5.10) regarded as a probability distribution on the set of non-negative integers.

Definition 5.10 (Poisson distribution). *Let $\lambda > 0$. We say that a discrete real-valued random variable X follows the $\text{Poisson}(\lambda)$ distribution, written $X \sim \text{Poisson}(\lambda)$, if its probability mass function is given by*

$$f_X(x) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!}, & \text{for } x \in \{0, 1, \dots\}; \\ 0, & \text{otherwise.} \end{cases} \quad (5.13)$$

We call λ the rate or mean parameter of the Poisson distribution.

Our probabilistic derivation of the Poisson distribution presents our first instance of a (distributional) *limit theorem* in this course.

Theorem 5.11 (Poisson distribution as a limit of binomial distribution). *Let $\lambda > 0$. Let $(\lambda_n)_{n=1}^{\infty}$ be a non-negative sequence with $\lim_{n \rightarrow \infty} \lambda_n = \lambda$, and define $p_n = \frac{\lambda_n}{n}$.⁴ For each n , let X_n be a $\text{Binomial}(n, p_n)$ random variable and let $f_n := f_{X_n}$ be its pmf. Then for $x \in \{0, 1, \dots\}$ we have*

$$\lim_{n \rightarrow \infty} f_n(x) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Thus the pmf of f_n converges for all x to the pmf of the Poisson distribution with rate parameter λ .

⁴Since $\lim_{n \rightarrow \infty} \lambda_n = \lambda$ we have $\lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} \frac{\lambda_n}{n} = 0$. Thus for n sufficiently large we have $0 \leq p_n < 1$ and so $\text{Binomial}(n, p_n)$ is well defined. (In this result we are only interested in the limit as $n \rightarrow \infty$.)

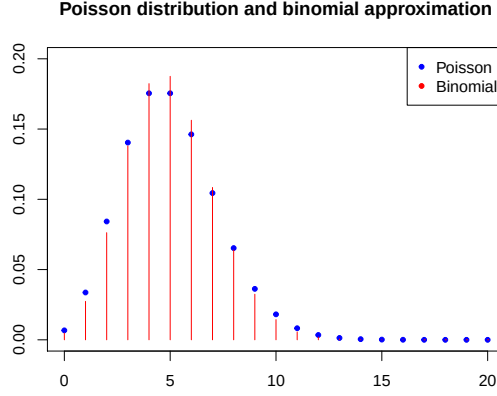


Figure 5.1: Binomial approximation to the Poisson distribution. We plot the pmfs of $\text{Poisson}(\lambda)$ and $\text{Binomial}(n, p_n)$, where $\lambda = 5$, $n = 40$ and $p_n = \frac{\lambda}{n}$. (The values for $x > 20$ are not shown as they are very small).

Thus the Poisson distribution arises (in the limit) in the number of successes in a large number of independent Bernoulli trials with a tiny success probability. Intuitively, this result suggests that “ $\lim_{n \rightarrow \infty} \text{Binomial}(n, p_n) = \text{Poisson}(\lambda)$ ” (at the level of the distributions but *not* the random variables) if $\lim_{n \rightarrow \infty} p_n n = \lambda$. Figure 5.1 we show that the approximation $\text{Poisson}(\lambda) \approx \text{Binomial}(n, p_n)$ works quite well even for moderate values of n .

Proof of Theorem 5.11. Fix $x \in \{0, 1, \dots\}$ and consider the binomial pmf

$$f_n(x) = \binom{n}{x} p_n^x (1 - p_n)^{n-x} = \binom{n}{x} \frac{\lambda_n^x}{n^x} \left(1 - \frac{\lambda_n}{n}\right)^{n-x}.$$

Using the definition (3.7) of the binomial coefficient, we may write

$$f_n(x) = \frac{1}{x!} \frac{n}{n} \frac{n-1}{n} \dots \frac{n-x+1}{n} \lambda_n^x \left(1 - \frac{\lambda_n}{n}\right)^n \left(1 - \frac{\lambda_n}{n}\right)^{-x}.$$

Let’s analyze the behaviour of the terms as $n \rightarrow \infty$:

- $\lim_{n \rightarrow \infty} \frac{n}{n} \frac{n-1}{n} \dots \frac{n-x+1}{n} = 1$ since x , the number of terms, is finite.
- $\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda_n}{n}\right)^n = e^{-\lambda}$ by (5.12).
- $\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda_n}{n}\right)^{-x} = 1$ since $\lim_{n \rightarrow \infty} \frac{\lambda_n}{n} = 0$ and x is finite.

It follows that $\lim_{n \rightarrow \infty} f_n(x) = \frac{\lambda^x e^{-\lambda}}{x!}$ and the theorem is proved. $\square$

We describe a situation which leads to the context of Theorem 5.11. Consider a random process which leads to “events” at isolated points in time. For example, consider *ionizing radiation* whose occurrences can be detected by a

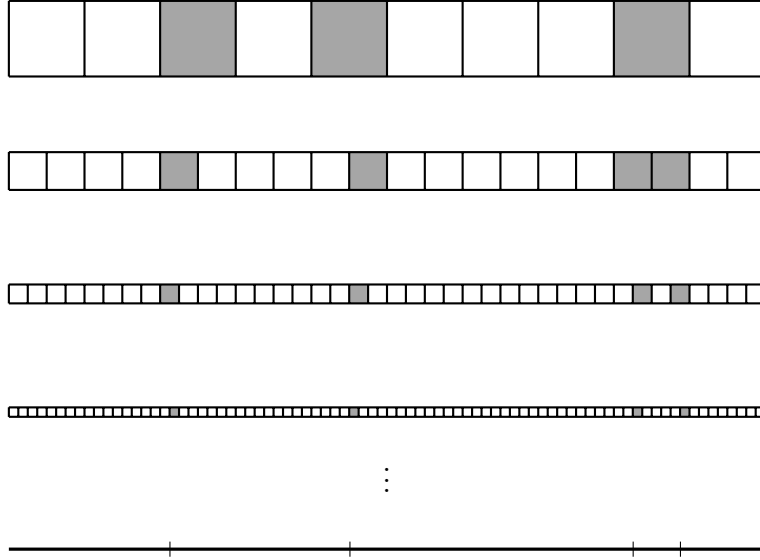


Figure 5.2: Modelling the occurrences of “point events” by discretizing a continuous interval into subintervals. A box is highlighted if an event (or more) occurs in the corresponding subinterval. In the derivation of the Poisson distribution the activities in disjoint subintervals are assumed to be independent from each other.

Geiger counter.⁵ Suppose we are interested in modelling the (random) total number of occurrences in a fixed time unit interval (e.g. 1 minute).

Suppose we divide the interval into n equal subintervals, e.g., due to the accuracy or processing limit of the measuring device. Using the device, we can tell whether an event (or more) occurs in each subinterval. So we have a binary outcome for each subinterval (see Figure 5.2). Suppose that the underlying random process satisfies the following (idealized) properties:

- The occurrence (or not) of events in disjoint time subintervals are independent.
- The “intensity” of occurrence is *constant* over time (this property is called *time homogeneity*). Then the probability of observing at least one event in a given subinterval is the same for all subintervals (with the same length).

Now the occurrences (or not) in the subintervals may be regarded as a sequence of i.i.d. Bernoulli trials. If p_n is the probability of occurrence in each subinterval, the total number is distributed as $\text{Binomial}(n, p_n)$.

Given n Bernoulli trials with success probability p_n , on average there are np_n total counts (this is the concept of *expectation* which will be introduced in Section 6). We may think of $\lambda_n = np_n$ as a measure of the “intensity” of the discretized process. Specifically, λ_n may be regarded as the *average*

⁵See https://en.wikipedia.org/wiki/Geiger_counter.

number of counts per unit time (since the whole interval represents unit time by assumption).

But the discretization, which leads to $\text{Binomial}(n, p_n)$, is only an approximation of an underlying process which takes place in continuous time. To find a suitable model for this process, we imagine letting $n \rightarrow \infty$. We expect that as $n \rightarrow \infty$, λ_n should tend to the intensity of the underlying process. So we suppose $\lambda = \lim_{n \rightarrow \infty} \lambda_n$ exists and is the average number of counts per unit time. Note that $p_n = \frac{\lambda_n}{n}$ becomes smaller and smaller as n increases. So there is a large number of i.i.d. Bernoulli trials with a tiny success probability. By Theorem 5.11, the total number of counts, if recorded exactly as they occur in continuous time, should be modelled by the $\text{Poisson}(\lambda)$ distribution.

Remark 5.12 (Foreshadowing a topic in future courses). In the above context, it is possible to model not only the total number of counts but also the *individual locations* of the point events. This is achieved by the *Poisson process* which is a fundamental stochastic process in probability theory. Its description involves the Poisson distribution and also the *exponential distribution* to be introduced in Section 5.5 below.

5.4 Hypergeometric distribution

As hinted in Exercise 2 in Section A.3.1, the *hypergeometric distribution* arises naturally in sampling without replacement.

Definition 5.13 (Hypergeometric distribution). *Let $N \geq 1$, $0 \leq K \leq N$ and $1 \leq n \leq N$ be integers. A discrete real-valued random variable X is said to follow the hypergeometric distribution $\text{Hypergeometric}(N, K, n)$ if its probability mass function is given by*

$$f_X(x) = \begin{cases} \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}, & \text{if } x \in \{0, \dots, K\}; \\ 0, & \text{otherwise.} \end{cases} \quad (5.14)$$

The word *hypergeometric* comes from a connection with the classical *hypergeometric function* whose precise definition is not relevant for our purposes.

Theorem 5.14. *Consider an urn with N balls such that $0 \leq K \leq N$ are red and the other $N - K$ balls are blue. Suppose n balls are randomly drawn from the urn without replacement. If X represents the number of red balls drawn, then $X \sim \text{Hypergeometric}(N, K, n)$.*

Proof. Since the urn contains only K red balls, it is clear that for $x > K$ we have $\mathbb{P}(X = x) = 0$. Now let $x \in \{0, \dots, K\}$ be the required number of red balls. By construction, each set of n balls is chosen with the same probability, namely $1/\binom{N}{n}$ (see Exercise 2 in Section A.3.1). There are $\binom{K}{x}$ ways to choose the x red balls and $\binom{N-K}{n-x}$ ways to choose the blue balls. It follows (by the product rule of counting) that

$$\mathbb{P}(X = x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}.$$

□

Remark 5.15. We may generalize the hypergeometric distribution to multiple categories. The resulting distribution is sometimes called the *multivariate hypergeometric distribution*. You may derive it as an exercise or look it up.

Sampling without replacement is generally more difficult to analyze since the results of the previous draws affect the probabilities of the future draws. Intuitively, if N (population size) is much larger than n (sample size), the effect of sampling without replacement becomes more and more negligible. In the limit (say when n is fixed and $N \rightarrow \infty$), one should recover sampling with replacement and hence the binomial distribution. This idea is made precise by the following theorem.

Theorem 5.16 (Binomial distribution as a limit of hypergeometric distribution). *Let $0 \leq p \leq 1$ and let $n \geq 1$ be an integer. For each $\ell \geq 1$, let N_ℓ and K_ℓ be positive integers satisfying the following conditions.*

$$0 \leq K_\ell \leq N_\ell, \quad \lim_{\ell \rightarrow \infty} \frac{K_\ell}{N_\ell} = p, \quad \lim_{\ell \rightarrow \infty} N_\ell = \lim_{\ell \rightarrow \infty} K_\ell = \infty.$$

For each ℓ , let f_ℓ be the pmf of $\text{Hypergeometric}(N_\ell, K_\ell, n)$. Then for $x \in \{0, 1, \dots, n\}$ we have

$$\lim_{\ell \rightarrow \infty} f_\ell(x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

Proof. See Exercise 4 of Section A.5.1. □

Thanks to this limit theorem, when studying randomized surveys one typically assumes independent sampling (provided that the sample size is small relative to the population size) even though strictly speaking the respondents are sampled without replacement.

5.5 Exponential distribution

Definition 5.17 (Exponential distribution). *Let $\lambda > 0$. A continuous real-valued random variable X is said to have the exponential distribution with rate parameter $\lambda > 0$ ($\text{Exp}(\lambda)$), written $X \sim \text{Exp}(\lambda)$, if its probability density function is given by*

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{for } x > 0; \\ 0, & \text{otherwise.} \end{cases} \quad (5.15)$$

The function f_X in (5.15) defines a probability density function since

$$\int_0^\infty \lambda e^{-\lambda x} dx = \lim_{x \rightarrow \infty} (1 - e^{-\lambda x}) = 1.$$

Here we used the fact that the antiderivative of $\lambda e^{-\lambda x}$ is $-e^{-\lambda x}$ (up to an additive constant). By a similar computation, we see that the cumulative distribution function of $\text{Exp}(\lambda)$ is given by

$$F_X(x) = \int_{-\infty}^x f_X(z) dz = \begin{cases} 0, & \text{for } x < 0; \\ 1 - e^{-\lambda x}, & \text{for } x \geq 0. \end{cases} \quad (5.16)$$

In particular, if $X \sim \text{Exp}(\lambda)$ its tail probability (analogous to (5.8) in Section A.5.1 for the geometric distribution) is given by

$$\mathbb{P}(X \geq x) = \mathbb{P}(X > x) = 1 - F_X(x) = e^{-\lambda x},^6 \quad (5.17)$$

which decays exponentially with rate λ .

The exponential distribution is frequently used to model *waiting times* such as the time between car accidents, arrival of a new customer, etc. We motivate it probabilistically by showing that the *exponential distribution is a continuous analogue of the geometric distribution*, thus giving another instance of a limit theorem. This is closely related to our probabilistic motivation of the Poisson distribution given in Section 5.3. Here, we also need to *scale* the random variables properly to obtain the limit.

Recall from Section 5.2 that the geometric distribution models the waiting time of the first success in an i.i.d. sequence of Bernoulli trials. Let's be more precise about the unit of time. Suppose first that we perform one Bernoulli trial per unit time (say minute). Then $X \sim \text{Geometric}(p)$ represents the (random) number of minutes to get the first success. Since the probability of success in each trial is p , on average we need to wait $\frac{1}{p}$ units of time. (This again corresponds to the concept of expectation.) Suppose now we perform n trials in unit time. Suppose also that the success probability p_n depends on n . Let $X_n \sim \text{Geometric}(p_n)$ be the number of trials required. Then the actual time of the first success is

$$T_n = \frac{1}{n} X_n, \quad (5.18)$$

since a trial takes $\frac{1}{n}$ unit of time. From the above discussion, on average we need to wait $\frac{1}{np_n}$ units of time. Let $\lambda_n = np_n$. For a limiting distribution to exist, we assume that $\lim_{n \rightarrow \infty} \lambda_n = \lambda > 0$ exists and is a positive constant. Again, we think of λ as the underlying *intensity*, so that $\frac{1}{\lambda}$ is the limiting “average waiting time”.

Consider now the *tail probability* $\mathbb{P}(T_n > t)$ for $t > 0$; this is the probability that we need to wait more than t units of time. Write $t = (tn)\frac{1}{n}$. So t units of time corresponds to about tn trials.⁷ It follows from the tail probability of the geometric distribution (see (5.8)) that

$$\mathbb{P}(T_n > t) = \mathbb{P}(X_n > nt) \approx (1 - p_n)^{nt} = \left(1 - \frac{\lambda_n}{n}\right)^{nt}.$$

⁶The first equality holds since for any continuous distribution we have $\mathbb{P}(X = x) = 0$.

⁷Note that tn is not necessarily an integer but we can round it to the nearest integer.

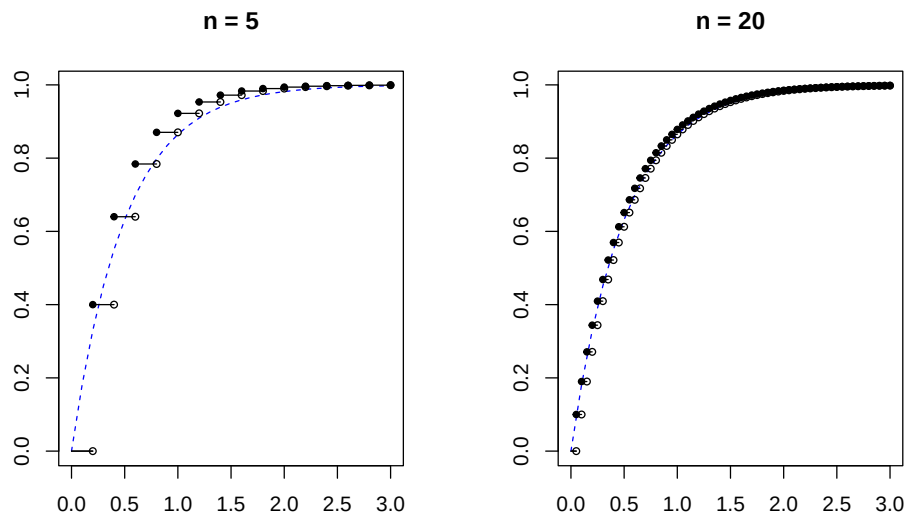


Figure 5.3: Approximating the exponential distribution in terms of a geometric distribution. We plot the cdf of $\text{Exp}(\lambda)$ with $\lambda = 2$ (blue, dashed). We also plot (in black) the cdf of the random variable $\frac{1}{n}X_n$ where $X_n \sim \text{Geometric}(p_n)$ and $p_n = \frac{\lambda}{n}$. Left: $n = 5$. Right: $n = 20$. It can be shown that as $n \rightarrow \infty$ the cdf of $\frac{1}{n}X_n$ converges pointwise to that of $\text{Exp}(\lambda)$.

From (5.12), the last expression converges as $n \rightarrow \infty$ to $e^{-\lambda t}$, which by (5.17) is the tail probability of the exponential distribution $\text{Exp}(\lambda)$. So, we have argued that as $n \rightarrow \infty$, the distribution of $T_n = \frac{1}{n}X_n$ which is discrete and where $X_n \sim \text{Geometric}(\frac{\lambda}{n})$, “converges” to $\text{Exp}(\lambda)$ which is continuous. See Figure 5.3 for an illustration in terms of the cumulative distribution function. This idea can and will be made precise with the concept of *convergence in distribution* which will be introduced later.

Before ending this section, we use the exponential distribution to illustrate a general way to think about continuous distributions on the real line. Namely, *a continuous random variable, with a prescribed distribution, can be obtained by transforming a uniform random variable by the (inverse of the) given cumulative distribution function*. Another example (the *Cauchy distribution*) is given in Exercise 7 in Section A.5.1.

Proposition 5.18 (Inverse transform method (special case for exponential distribution)). *Let $U \sim \text{Uniform}(0, 1)$ be a uniform random variable. Given $\lambda > 0$, let $F(x) = 1 - e^{-\lambda x}$, $x > 0$, be the cumulative distribution function of $\text{Exp}(\lambda)$ on $(0, \infty)$ (see (5.16)). Let*

$$F^{-1}(u) = \frac{-1}{\lambda} \log(1 - u), \quad u \in (0, 1), \quad (5.19)$$

be the inverse of F which maps $(0, 1)$ onto $(0, \infty)$.⁸ Define a random variable X by $X = F^{-1}(U)$. Then $X \sim \text{Exp}(\lambda)$.

⁸The inverse $F^{-1}(u)$ is defined by the solution x to the equation $F(x) = u$. Given $u \in (0, 1)$,

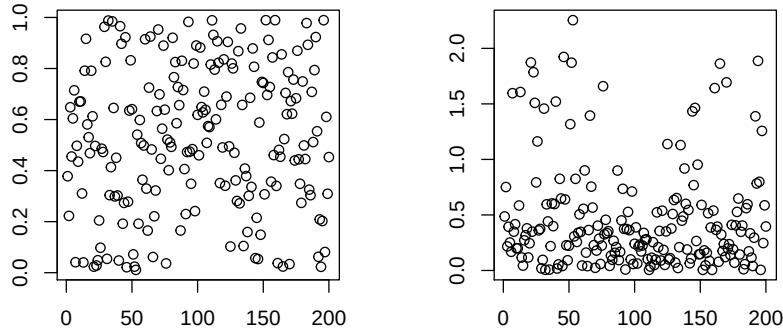


Figure 5.4: Left: 200 independent samples from the uniform distribution (obtained by the `runif()` function in R). Right: Samples from $\text{Exp}(\lambda)$ obtained by $X_i = -\frac{1}{\lambda} \log U_i$, where $\lambda = 2$.

Proof. Since U takes values in $(0, 1)$ and $F^{-1} : (0, 1) \rightarrow (0, \infty)$, $X = F^{-1}(U)$ takes values in $(0, \infty)$.⁹ We check that the cumulative distribution of X is F .

Fix $x > 0$. Since F is a strictly increasing bijection from $(0, \infty)$ to $(0, 1)$, we have $\{X \leq x\} = \{F^{-1}(U) \leq x\} = \{U \leq F(x)\}$. Hence

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x),$$

where in the last equality we used the fact that the cumulative distribution function of $\text{Uniform}(0, 1)$ is the identity on $(0, 1)$ (see Example 4.19). Since the cumulative distribution function of X is identical to that of $\text{Exp}(\lambda)$, we have $X \sim \text{Exp}(\lambda)$ (see Remark 4.11). $\square$

The *inverse transform method*, a special case of which is illustrated in Proposition 5.18, gives a way to generate samples from a given univariate distribution using samples from the uniform distribution. Results such as the above one are important for numerical implementation since by suitably transforming samples from the uniform distribution (or some reference distribution) we can obtain (exact or approximate) samples from more complicated distribution. In practice, (approximate) samples from the uniform distribution can be obtained from *pseudo-random number generators*. For the exponential distribution, here is a specific algorithm. Let $U_1, U_2, \dots$ be an i.i.d. sequence of $\mathcal{U}(0, 1)$ random variables. Then $-\frac{1}{\lambda} \log U_1, -\frac{1}{\lambda} \log U_2, \dots$ is a sequence of i.i.d. $\text{Exp}(\lambda)$ random variables.¹⁰ In this particular case, using $-\frac{1}{\lambda} \log U_i$ instead of $-\frac{1}{\lambda} \log(1 - U_i)$ as in (5.19) saves the operation of computing $1 - U_i$ since $1 - U_i$ is also uniformly distributed on $(0, 1)$. See Figure 5.4 for an illustration where the rate parameter is $\lambda = 2$.

solving $1 - e^{-\lambda x} = u$ gives $x = -\frac{1}{\lambda} \log(1 - u)$.

⁹For readability we omit technical qualifiers such as “with probability 1.”

¹⁰That the samples $-\frac{1}{\lambda} \log U_i$ are independent follows from Exercise 5(c) in Section A.4.1.